

ASHUTOSH TRIPATHI

B.Tech Computer Science & Engineering · 2nd Semester

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PROFESSIONAL OBJECTIVE

Highly motivated 2nd-semester B.Tech CSE student at IIIT Manipur with a strong foundation in C programming and Data Structures. Passionate about Machine Learning, Data Science, and software development. Seeking internships and collaborative opportunities to apply algorithmic problem-solving and deep learning concepts to real-world challenges.

EDUCATION

Indian Institute of Information Technology (IIIT), Manipur

B.Tech in Computer Science & Engineering | 2025 – Present

Current Status: Pursuing (2nd Semester) · Roll No.: 250101017

CBSE Board

Class XII – Science (PCM) | 2025

Percentage: 84%

UP Board

Class X | 2023

Percentage: 87%

TECHNICAL SKILLS

Languages:	C (Proficient), Python (Intermediate)
DSA:	Arrays, Linked Lists, Stacks, Queues, Trees (Implemented in C)
ML & Data Science:	Basics of Deep Learning, LSTM Networks, Time-Series Forecasting, Data Preprocessing (NumPy, Pandas)
Tools & IDEs:	VS Code, Git/GitHub, Jupyter Notebook

ACADEMIC PROJECTS & RESEARCH

SmartAir: Predictive Modeling for Mine Safety

Academic Research / Self-Study

- Conducted an extensive literature review of 5+ top-tier research papers (IEEE, MDPI) on applying Deep Learning to toxic gas detection.
- Designed a conceptual framework for using Long Short-Term Memory (LSTM) neural networks to predict Carbon Monoxide (CO) and Methane (CH₄) concentrations in time-series data.
- Analyzed data preprocessing techniques, sliding window mechanisms, and feature engineering for optimizing ML models on multi-sensor datasets.

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Programming Fundamentals
- Mathematics for Computer Science
- Digital Electronics
- Introduction to Artificial Intelligence

SOFT SKILLS & ABILITIES

- **Analytical Thinking:** Strong mathematical foundation and problem-solving mindset developed through DSA.
- **Research-Oriented:** Ability to read, comprehend, and extract data from complex scientific research papers.
- **Adaptability:** Fast learner, actively expanding skill set from core C programming to Python-based Machine Learning.

PERSONAL DETAILS

Date of Birth:	09 July 2007	Nationality:	Indian
Languages:	English (Professional Proficiency), Hindi (Native)		